Multi-Armed Bandits with Arriving Arms: Sequential Screening, Dynamic Regret, and Sublinear Guarantees

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Abstract

We study a stochastic multi-armed bandit problem in which the set of available arms expands over time. This setting arises in sequential experimentation when new actions or treatments become available during an ongoing study, making regret against a single best arm in hindsight inappropriate. We instead evaluate performance relative to the best arm currently available, leading to a dynamic-regret criterion for arriving-arm environments. To address the resulting challenges of arrival information discrepancy (AID) and a drifting benchmark (DB), we propose UCB for Arriving Arms (UCB-AA), an elimination-based procedure with an aiding preliminary screening step for newly arrived arms before full competition with incumbent arms. We show that UCB-AA attains regret bounds that depend explicitly on the arrival process, achieves sublinear dynamic regret under regularity conditions on gap evolution, and admits an online extension for unknown horizons. Simulation results show that UCB-AA reduces wasted pulls and maintains a smaller active arm set while preserving competitive regret performance.

Keywords: Sequential experimentation; adaptive screening; expanding action sets; stochastic bandits; arrival information discrepancy; drifting benchmark

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1 Introduction

The classical multi-armed bandit framework assumes that the set of available actions is fixed in advance. In many modern applications, however, the action set expands over time. Precision oncology provides a representative example: as genomic profiling has advanced, the set of approved targeted therapies for non-small cell lung cancer has grown substantially, so a trial designed around a fixed collection of treatments may become outdated before it is completed. Similar issues arise in online advertising and other adaptive decision problems in which new options appear sequentially.

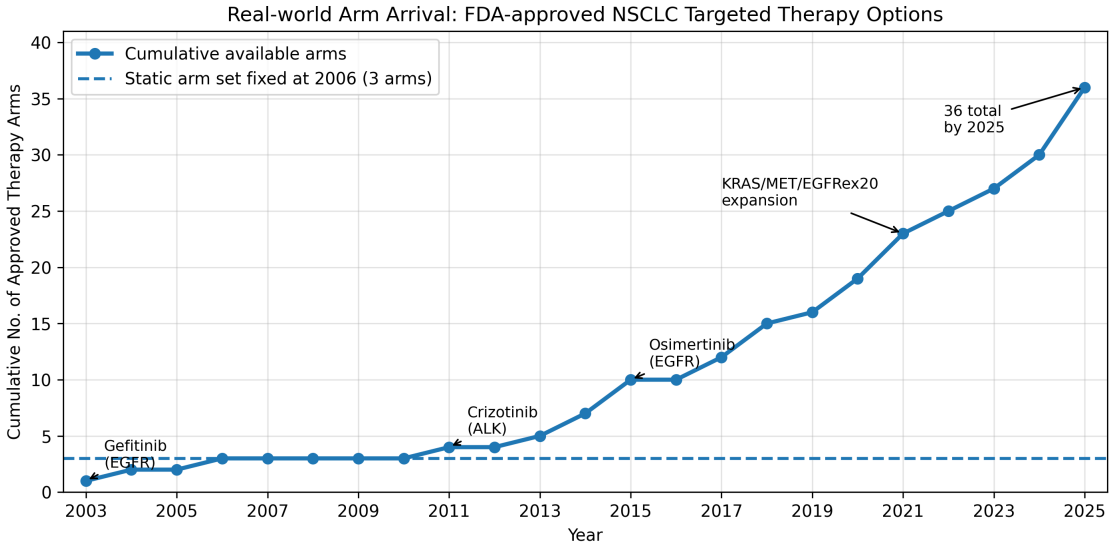


Figure 1: **The rapid expansion of the decision space in precision oncology.** FDA approval data for NSCLC (2003–2025). The solid line shows the cumulative number of targeted and immunotherapies (arms), from 1 in 2003 (Gefitinib) to 36 by 2025. A static algorithm fixed to the 2006 arm set would fail to incorporate most subsequently available treatments, illustrating the need for methods that accommodate arriving arms.

Motivated by such problems, we study the **multi-armed bandit problem with arriving arms** (MAB-AA). Let A_t denote the set of arms available at time t, and suppose that $A_t \subseteq A_{t+1}$, so that new arms arrive over time and remain available thereafter. In this setting, performance is more naturally evaluated against the best arm currently available than against the single best arm in hindsight, since the latter may not have been available earlier.

Relative to the classical stochastic bandit setting, MAB-AA introduces two main difficulties. The first is *arrival information discrepancy (AID)*: newly arrived arms must be explored, which repeatedly diverts samples away from well-understood incumbent arms. The second is a *drifting benchmark (DB)*: as better arms enter the system, both the identity of the optimal arm and the regret benchmark may change over time.

The arriving-arm setting is related to, but distinct from, several existing bandit formulations. Unlike infinite-armed bandits (Berry, Chen, et al., 1997), where arms are actively sampled from a reservoir, new arms here arrive exogenously and remain available thereafter. Unlike mortal bandits

(Chakrabarti et al., 2008), the defining feature is not arm death, and unlike sleeping bandits (Kleinberg, Niculescu-Mizil, and Sharma, 2010), the available set evolves monotonically over time. Our setting is also related to bandits with moving comparators: in nonstationary and switching bandits, the best arm may change because reward distributions vary over time (Besbes, Gur, and Zeevi, 2014; Auer, Gajane, and Ortner, 2019), whereas in MAB-AA each arm has a stationary reward law after arrival and benchmark changes are driven only by the exogenous arrival of new arms.

Most closely related, Ghalme et al. (2021) studies ballooning bandits with regret measured against the best currently available arm, but focuses on feasibility of sublinear regret under restricted arrival patterns. Relatedly, Qi et al. (2025) considers an expanding-arm extension in graph-feedback bandits with similar arms. In contrast, our focus is the unstructured arriving-arm case, which is more suitable in applications where new options are introduced over time, previously introduced options remain available, and reliable similarity or contextual structure across options is unavailable. Examples include online product or content recommendation with continuously added items, adaptive A/B testing with newly launched treatments or designs, and sequential decision problems in which candidate actions are generated by an external pipeline rather than chosen by the learner.

From a statistical perspective, multi-armed bandit problems have long been studied as models for sequential allocation and adaptive experimentation, dating back to Robbins (1952) and the asymptotically efficient allocation theory of Lai and Robbins (1985); see also Lai (1987) for connections to adaptive treatment allocation. This viewpoint has motivated a broad statistical literature, including Bayesian and index-based approaches to sequential allocation (Gittins, 1979; Berry and Fristedt, 1985), response-adaptive randomization in clinical trials (Zelen, 1969; Wei and Durham, 1978; Villar, Wason, and Bowden, 2015; Williamson and Villar, 2020; Aziz, Kaufmann, and Riviere, 2021), covariate-dependent and nonparametric bandit methods (Woodroofe, 1979; Yang and Zhu, 2002; Rigollet and Zeevi, 2010; Perchet and Rigollet, 2013), and recent high-dimensional, semiparametric, and transfer-learning formulations (Qian and Yang, 2016; Qian, Ing, and Liu, 2024; Cai, Cai, and Li, 2024). These works share with ours the broad goal of learning from accumulated evidence to reduce allocation to inferior alternatives, but they focus on a fixed set of treatments or actions, and thus are not applicable to the arising arm setting that we study.

Methodologically, the two challenges call for different but complementary design principles. To address AID, our approach combines elimination steps (Even-Dar, Mannor, and Mansour, 2006; Auer and Ortner, 2010; Perchet and Rigollet, 2013; Qian and Yang, 2016) with a pre-elimination mechanism tailored to the expanding action set. The key idea is to prevent the main competition stage from being repeatedly disrupted by every newly arriving arm. Instead, late arrivals are first screened against incumbent candidates, so that clearly inferior arms can be removed before receiving the full exploration budget assigned to serious contenders. To handle DB, the algorithm repeatedly updates the relevant comparison set as new arms arrive, and the analysis evaluates regret against the best arm currently available rather than against a fixed hindsight oracle.

Based on these ideas, we propose **UCB for Arriving Arms (UCB-AA)**. For UCB-AA, we derive regret upper bounds that depend explicitly on the arrival process, establish sublinear dynamic

regret under suitable conditions on gap evolution, study representative lower-bound regimes, and develop an online extension for unknown horizons.

Our contributions are threefold. First, we formulate the arriving-arm stochastic bandit problem using a dynamic-regret benchmark relative to the best currently available arm. Second, we propose a screening-based elimination procedure that is tailored to the information imbalance between incumbent and newly arrived arms. Third, we provide regret guarantees, lower-bound insights in representative regimes, and an online extension, together with simulations showing that UCB-AA reduces wasted pulls and maintains a compact active arm set.

The remainder of the paper is organized as follows. Section 2 formulates the problem and introduces the regret criterion. Section 3 presents UCB-AA, its theoretical guarantees, and an online extension for unknown horizons. Section 4 develops lower bounds in representative regimes. Section 5 reports numerical results. Section 6 concludes.

2 Multi-Armed Bandit Problem with Arriving Arms

2.1 Problem Formulation

We consider a sequential decision process over time steps $t = 1, 2, \ldots$. Let $\mathcal{A}$ denote the (possibly infinite) set of all arms. At each time step t, only a subset $\mathcal{A}_t \subset \mathcal{A}$ is available, and we assume that the available set is non-decreasing: $\mathcal{A}_t \subseteq \mathcal{A}_{t+1}$. For any arm $i \in \mathcal{A}$, define its arrival time as $\tau_i := \inf\{t \geq 1 : i \in \mathcal{A}_t\}$. Each arm i is associated with an unknown reward distribution on $[0, 1]$ with mean r_i. When arm i is pulled at time t, a reward $X_{t,i}$ is generated; for each fixed i, we assume $\{X_{t,i}\}_{t\geq 1}$ are i.i.d. with mean r_i, and rewards are independent across arms. The arrival process is treated as exogenous to the learner.

A policy is a sequence $\pi = (\pi_t)_{t\geq 1}$, where each decision rule π_t maps the past history to an available arm $I_t := \pi_t(H_{t-1}) \in \mathcal{A}_t$, where $H_{t-1} := (\mathcal{A}_s, I_s, X_{s,I_s})_{s=1}^{t-1}$ denote the observed history up to time $t - 1$. Then the learner observes only X_{t,I_t}. Because the available set changes over time, the best arm may also change. We define the best mean reward at time t as $r^*(t) := \max_{i\in\mathcal{A}_t} r_i$.

For any (possibly random) stopping time τ, define the cumulative regret by the (expected) regret against the best available arm at each time:

$$R_\tau(\pi) := \sum_{t=1}^{\tau} (r^*(t) - r_{I_t}).$$

For a deterministic horizon T, this reduces to $R_T(\pi)$. For the round-based procedure analyzed below, the relevant stopping time is $T_0(\pi)$, the total number of pulls used to complete M rounds. This benchmark differs from the usual regret against the single best arm in hindsight. In the arriving-arm setting, the terminal best arm in $\mathcal{A}_T$ may not have been available in earlier periods, so comparing the learner with that arm throughout the entire horizon may overstate the achievable performance. The criterion above instead compares the learner at each time t with the best arm that was actually available at that time. When $\mathcal{A}_t$ is constant over time, this formulation reduces to the classical stochastic multi-armed bandit problem.

2.2 Main Statistical Challenges

Relative to the classical fixed-arm stochastic bandit model, the arriving-arm setting introduces two related sources of difficulty.

Arrival Information Discrepancy (AID): A first difficulty is arrival information discrepancy. With a finite sampling budget up to time T, the learner must allocate samples both to incumbent arms and to newly arriving arms whose rewards are still highly uncertain. Under continual arrivals, this repeated re-exploration can prevent the learner from concentrating pulls on the best currently available arm. Consequently, without additional regularity conditions on the arrival process, sublinear regret may be unattainable.

Drifting Benchmark (DB): In contrast to the classical setting where the best mean reward is fixed, the benchmark in MAB-AA is time-dependent: $r^*(t) = \max_{i\in\mathcal{A}_t} r_i$. Since $\mathcal{A}_t$ expands over time, both the identity of the best arm and the value $r^*(t)$ may change when new arms arrive. Consequently, for a fixed arm i, the suboptimal gap $r^*(t) - r_i$ is generally time-varying (and non-decreasing in t). This drifting benchmark complicates both algorithm design and analysis, because the learner must continuously compare well-sampled incumbent arms with newly arriving arms whose estimates are still highly uncertain.

3 UCB for Arriving Arms (UCB-AA) Algorithm

3.1 Algorithm Description

UCB-AA is an elimination-based procedure organized by rounds. Its main idea is simple: newly arriving arms are not allowed to compete immediately with well-sampled incumbent arms. Instead, they are first buffered and screened at lower cost. Only those that survive this screening step enter full competition in the next stage. This design aims to reduce wasted exploration on clearly suboptimal late arrivals while preserving the ability to detect genuinely competitive new arms.

Fix a finite number of rounds M. For each round $m = 0, 1, \ldots, M$, let A_m denote the set of arms arriving during round $m-1$, with A_0 the initial arm set, and define $A(m) := \bigcup_{j=0}^{m} A_j$. Let C_m denote the set of arms surviving round m. Then round m starts from the surviving incumbent arms C_{m-1} together with the newly admitted batch A_m. We write B_m for the active set in that round.

We assume that the cumulative number of arrived arms is bounded by a known nondecreasing function $K(m)$:

$$|A(m)| \le K(m), \qquad |A_{m+1}| \le K(m+1) - K(m).$$

For simplicity, assume that the best arm is unique in each round. Let $r_{*(m)} := \max_{i\in B_m} r_i$, and let $*(m) \in B_m$ denote the corresponding optimal arm. For each arm i, define its arrival round $t_i := \min\{m : i \in A_m\}$, and its round-$m$ suboptimality gap

$$\Delta_i(m) := \begin{cases} r_{*(m)} - r_i, & m \ge t_i, \\ 0, & m < t_i. \end{cases}$$

In round m, UCB-AA proceeds in two stages. As shown in Algorithm 1, the earlier phases $p = 0, \ldots, m-2$ are used for pre-elimination of newly arrived arms, while the last two phases $p = m-1, m$ are used for elimination on the merged active set.

Algorithm 1 UCB for Arriving Arms (UCB-AA)

Input: number of rounds M, parameter $N > 0$.

Initialization: $\tilde{\Delta}_0 := 1$, $B_{-1} := \varnothing$, $C_{-1} := \varnothing$, $T_0 \leftarrow 0$.
procedure ELIM(S, R, p)
  $n_p := \left\lceil \frac{2\log(N\tilde{\Delta}_p^2)}{\tilde{\Delta}_p^2} \right\rceil$
  Pull each $j \in S$ to n_p total pulls; update T_0 after each pull
  Let $b_p := \sqrt{\frac{\log(N\tilde{\Delta}_p^2)}{2n_p}}$; set $\mathrm{LCB}_j = \hat{r}_j(n_p) - b_p$, $\mathrm{UCB}_j = \hat{r}_j(n_p) + b_p$
  Remove from S any j with $\mathrm{UCB}_j \le \max_{i \in R} \mathrm{LCB}_i$
  return S
end procedure
for $m = 0, 1, \ldots, M$ **do**
  Receive A_m and set $B_m^0 := C_{m-1} \cup A_m$
  if $|A_m| \neq 0$ and $m \ge 2$ **then**
    $A_m^0 := A_m$
    for $p = 0, 1, \ldots, m-2$ **do**
      $A_m^{p+1} \leftarrow \mathrm{ELIM}(A_m^p,\ B_m^p,\ p)$; $B_m^{p+1} := C_{m-1} \cup A_m^{p+1}$
    end for
  end if
  if $m = 0$ **then**
    $B_m^1 \leftarrow \mathrm{ELIM}(B_m^0,\ B_m^0,\ 0)$
  else
    for $p = m-1, m$ **do**
      $B_m^{p+1} \leftarrow \mathrm{ELIM}(B_m^p,\ B_m^p,\ p)$
    end for
  end if
  $C_m := B_m^{m+1}$, $\tilde{\Delta}_{m+1} = \frac{1}{2}\tilde{\Delta}_m$
end for

3.1.1 Stage I: Pre-elimination of newly arrived arms

At the beginning of round m, the newly admitted batch A_m is screened against the surviving incumbent set C_{m-1}. During the preliminary phases $p = 0, \ldots, m-2$, only arms in A_m are subject to removal: each surviving new arm is sampled to the phase-specific target n_p, and is discarded if its upper confidence bound falls below the largest lower confidence bound of the reference set. This stage exploits the information asymmetry between incumbents and arrivals. Incumbent arms have typically accumulated more samples, whereas newly arrived arms are still highly uncertain. Screening arrivals before full competition therefore prevents clearly inferior late arrivals from consuming the exploration budget assigned to serious contenders.

3.1.2 Stage II: Formal elimination on the active set

After pre-elimination, the surviving new arms are merged with the incumbent arms to form the active set for round m. UCB-AA then applies elimination to the full active set. At this stage, both incumbent and newly arrived arms may be removed. The surviving set is denoted by C_m and is carried into the next round. Thus, unlike a naive elimination rule, UCB-AA does not allow newly arrived arms to enter the main elimination stage immediately upon arrival. They must first pass the preliminary screening step in phases $p = 0, \ldots, m-2$. Once merged into the active set, however, all surviving arms are handled in the same way.

3.2 Regret Decomposition

Our analysis is indexed by the number of rounds M, rather than by fixing the total number of pulls in advance. We assume that the sampling budget is sufficient for UCB-AA to complete all M rounds, so the total number of pulls used up to the end of round M is a policy-dependent random variable. For a policy π, let $T_0(\pi)$ denote this total, and write

$$T := \mathbb{E}[T_0(\pi)].$$

Let $N_{i,m}^{\pi}$ be the number of times arm i is pulled during round m. Then $T_0(\pi) = \sum_{m=0}^{M} \sum_{i \in B_m} N_{i,m}^{\pi}$, where B_m is the active set at the beginning of round m.

Because UCB-AA updates its active set only at round boundaries, it is convenient to first analyze regret against the benchmark fixed at the beginning of each round. Define the *roundwise regret*

$$R_M(\pi) := \sum_{m=0}^{M} \sum_{i \in B_m} N_{i,m}^{\pi} \Delta_i(m).$$

This quantity charges each pull in round m against the best arm in the active set at the start of that round. Our target criterion, however, is the original dynamic regret

$$R_{T_0}(\pi) = \sum_{t=1}^{T_0(\pi)} \big(r^*(t) - r_{I_t}\big),$$

where $r^*(t) = \max_{i \in A_t} r_i$ is the best mean reward among all arms available at time t. These two regret notions differ because a benchmark-improving arm may arrive within a round but is not incorporated into the active set until the next round.

Accordingly, we decompose the dynamic regret as $R_{T_0}(\pi) = R_M(\pi) + D_M(\pi)$, where

$$D_M(\pi) := R_{T_0}(\pi) - R_M(\pi)$$

is the *delay regret*. The term $R_M(\pi)$ captures the main statistical cost of identifying competitive arms under the round structure, while $D_M(\pi)$ captures the additional loss caused by delayed incorporation of within-round benchmark improvements. The remainder of this section controls these two terms separately: we first bound the roundwise regret $R_M(\pi)$, and then bound the delay regret $D_M(\pi)$.

3.3 Upper Bound for the Roundwise Regret

We now bound the principal term in the decomposition of Section 3.2. The main question is how much regret is incurred before a suboptimal arm is removed under the elimination schedule of UCB-AA. To formalize this, define

$$m_i := \min\Big\{m \in \{t_i, t_i+1, \dots, M\} : \tilde{\Delta}_m < \tfrac{1}{2}\Delta_i(m)\Big\},$$

the first round in which arm i becomes statistically eliminable under the target resolution sequence. Also define

$$\tilde{m}_i := \min\Big\{m \in \{0, 1, \dots, M\} : \tilde{\Delta}_m < \tfrac{1}{2}\Delta_i(m_i)\Big\},$$

so that $\tilde{m}_i \le m_i$. We classify suboptimal arms according to the phase in which they are eliminated with high probability:

- G_1 $(m_i = t_i)$: arms eliminated when they first appear;
- G_2 $(m_i > t_i,\ \tilde{m}_i = m_i)$: arms are eliminated in the formal elimination phase $p = m_i$ of round m_i;
- G_3 $(m_i > t_i,\ \tilde{m}_i < m_i)$: arms are eliminated in the formal elimination phase $p = m_i - 1$ of round m_i.

For a threshold $\lambda \ge 2^{-M+1}$, define $A'(\lambda) := \{i \in A(M) : \Delta_i(M) > \lambda\}$. Then $A'(\lambda)$ admits the partition $A'(\lambda) = G_1(\lambda) \cup G_2(\lambda) \cup G_3(\lambda)$. Moreover, define

$$\bar{m}_i := \begin{cases} \tilde{m}_i, & i \in G_1(\lambda) \cup G_2(\lambda), \\ m_i - 1, & i \in G_3(\lambda). \end{cases}$$

Thus $\bar{m}_i$ is the elimination scale at which arm i is removed with high probability.

Theorem 1 (Upper bound for the roundwise regret)**.** *For any $\lambda \ge 2^{-M+1}$, the regret of UCB-AA up to round M satisfies*

$$\mathbb{E}R_M(\pi) \le \sum_{i:\Delta_i(M)\ge\lambda} \left(\tilde{S}_i + \frac{4(K(m_i)+5)\, R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \right) + T \max_{i:\Delta_i(M)\le\lambda} \Delta_i(M),$$

where $T = \mathbb{E}[T_0(\pi)]$ denotes the expected total number of pulls,

$$R_i := n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{M} (n_m - n_{m-1})\Delta_i(m),$$

and

$$\tilde{S}_i := \begin{cases} n_{\tilde{m}_i}\, \Delta_i(t_i), & i \in G_1(\lambda), \\ n_{t_i}\Delta_i(t_i) + \sum_{m=t_i+1}^{\bar{m}_i} (n_m - n_{m-1})\Delta_i(m), & i \in G_2(\lambda) \cup G_3(\lambda). \end{cases}$$

Theorem 1 decomposes the roundwise regret into three statistically interpretable components. The leading term, $\tilde{S}_i$, is the identification cost of certifying that a suboptimal arm should be removed at the resolution dictated by the elimination schedule. The second term quantifies the additional regret incurred on low-probability events where statistical fluctuations delay elimination or allow an incorrect survivor to persist. The final truncation term collects arms whose terminal gaps are below the target resolution λ, and may therefore be treated as effectively indistinguishable at the scale of the analysis.

To obtain a simpler rate statement, we impose the following stability condition.

Assumption 1 (Gap stability)**.** *For every suboptimal arm $i \in A'(\lambda)$, there exists a universal constant $\tilde{C} > 0$, independent of λ, such that*

$$\Delta_i(M) \le \tilde{C}\, \Delta_i(m_i).$$

Assumption 1 does not restrict the arrival times of arms; rather, it limits how much the benchmark can improve after an arm has already become statistically distinguishable. Equivalently, the post-detection benchmark increase cannot be larger than a constant multiple of the gap already visible at m_i. This is a mild condition in several common regimes. It holds with $\tilde{C} = 1$ if the best arm available at the terminal round has already arrived by m_i, or more generally if no later arrival improves the benchmark. It also holds whenever later improvements are moderate relative to the current detectable gap, for example if $r_{*(M)} - r_{*(m_i)} \le c\,\Delta_i(m_i)$ for a constant c, in which case $\tilde{C} = 1 + c$. This covers environments with bounded multiplicative benchmark growth after detection, as well as settings in which only a finite number of post-detection benchmark-improving arrivals occur and their cumulative improvement is comparable to the detection scale. Another simple case is a separated-gap regime: if every arm in $A'(\lambda)$ has $\Delta_i(m_i) \ge \delta_0 > 0$ for a fixed δ_0, then bounded rewards imply $\Delta_i(M) \le 1 \le \delta_0^{-1}\Delta_i(m_i)$. Thus the assumption mainly rules out extreme post-detection drift, where an arm is already clearly inferior but many substantially better arms arrive later and inflate its terminal gap by an unbounded factor.

Corollary 1. *Suppose that Assumption 1 holds for all $i \in A'(\lambda) = \{i \in A(M) : \Delta_i(M) > \lambda\}$. Let $K := K(M)$ and choose*

$$N = 10K\, n_M, \qquad \lambda \asymp \sqrt{\frac{K \log K + K}{T}}.$$

Then the roundwise regret of UCB-AA satisfies

$$\mathbb{E}R_M(\pi) \lesssim \sqrt{K \log K \cdot T}.$$

Corollary 1 shows that UCB-AA achieves sublinear roundwise regret when the effective number of arriving competitors remains moderate relative to the available sample size. Here $K = K(M)$ is the cumulative number of arms that may have appeared by round M, rather than the total size of the global arm universe. In particular, the bound is informative in regimes where $K(M) = o\left(\frac{T}{\log T}\right)$, so that the AID effect created by continual arrivals remains controllable.

Remark 1. *The parameter N must be fixed before running UCB-AA. In the theoretical analysis, we use the conservative calibration $N = 10K(M)n_M$, $n_M = \left\lceil \frac{2\log(N\widetilde{\Delta}_M^2)}{\widetilde{\Delta}_M^2} \right\rceil$, where $K(M)$ is a*

prior upper bound on the cumulative number of arrived arms. Although this implicit relation can be solved approximately, its role is mainly analytical: it ensures uniform control of low-probability failure events over all arms, but may lead to overly large pull targets n_m, and hence excessive forced exploration in both the pre-elimination and formal elimination stages. For implementation, we use a less conservative calibration based only on the monotonicity of the target sequence $\{n_m\}_{m=0}^{M}$. Since $f(x) = \frac{\log(Nx)}{x}$ is decreasing on $[\widetilde{\Delta}_M^2, 1]$ whenever $N\widetilde{\Delta}_M^2 \geq e$, a natural lower-end choice is $N_{\min} = \frac{e}{\widetilde{\Delta}_M^2}$. This optimistic calibration is not used in the regret proof, but provides a practical baseline with substantially smaller pull targets, and is adopted in the simulations.

3.4 Control of the Delay Regret

Delay regret arises because UCB-AA updates its active set only at round boundaries. If a benchmark-improving arm arrives in the middle of round k, then the true dynamic benchmark improves immediately on the original timeline, whereas the roundwise benchmark used in $R_M(\pi)$ is updated only at the beginning of round $k+1$. Thus $D_M(\pi)$ captures the additional loss caused by this deferred incorporation of within-round benchmark improvements.

Fix a round k. Suppose that within this round there are q_k benchmark-improving arrival events, with increments

$$\beta_{k,1}, \ldots, \beta_{k,q_k} \geq 0, \qquad \sum_{j=1}^{q_k} \beta_{k,j} =: \delta_k,$$

where $\delta_k = r_{*(k+1)} - r_{*(k)}$ is the total benchmark increase associated with round k. Let L_k be the total number of pulls in round k, and let $1 \leq \tau_{k,1} < \cdots < \tau_{k,q_k} \leq L_k$ denote the within-round locations of these improving arrivals. We define the exposure factor

$$\rho_k := \frac{1}{L_k \delta_k} \sum_{j=1}^{q_k} (L_k - \tau_{k,j} + 1)\beta_{k,j}, \qquad \delta_k > 0,$$

so that $0 \leq \rho_k \leq 1$. The factor ρ_k measures how much of round k is exposed to the benchmark increase: improvements that occur early contribute more to delay regret than those that occur late. For $\ell \geq k+1$, define the cumulative benchmark improvement after round k by

$$\Delta_{*(k)}(\ell) := r_{*(\ell)} - r_{*(k)} = \sum_{i=k}^{\ell-1} \delta_i.$$

We also define $m_{*(k)} = \min\Big\{\ell \geq k+1 : \tilde{\Delta}_\ell < \frac{1}{2}\Delta_{*(k)}(\ell)\Big\}$, the first round at which the cumulative improvement after round k becomes statistically detectable under the elimination schedule.

Remark 2. *For analytical convenience, we take the analyzed horizon to end at a checkpoint immediately after the last batch of arrivals has been admitted. Equivalently, one may append a terminal bookkeeping round with no further sampling. This convention does not require the learner to determine whether the final arrivals improve the benchmark; it only ensures that no arrivals remain unincorporated at the endpoint. Under this convention, the terminal delay contribution is null, so the endpoint condition involving $\rho_M \delta_M$ becomes vacuous.*

To bound $D_M(\pi)$, we impose a condition that limits how aggressively benchmark improvements can accumulate before they become statistically detectable. This condition is used to close the upper bound and should be viewed as a sufficient condition, not a necessary one.

Assumption 2 (Cumulative jump control)**.** *Assume that for every round k satisfying $\rho_k\delta_k > \lambda$, one has $m_{*(k)} > k+1$, and at least one of the following conditions holds:*

$$\Delta_{*(k)}(m_{*(k)}) \le \tilde{\Delta}_k \sqrt{\frac{M - m_{*(k)} + 1}{3\rho_k K(k)(M-k+1)}}, \tag{1}$$

$$\frac{\delta_k}{\Delta_{*(k)}(m_{*(k)}-1)} \le \frac{2^{m_{*(k)}-k-2}}{\rho_k K(k)}. \tag{2}$$

Assumption 2 is most naturally satisfied when benchmark improvements remain moderate until they become detectable at the current resolution scale. This includes, for example, settings in which improvements occur gradually, settings in which large improvements are rare, or settings in which most benchmark-improving arrivals occur late within a round and therefore have small exposure factors ρ_k.

Proposition 1 (Delay regret under cumulative jump control)**.** *Let $\lambda \ge 2^{-M+1}$. Under Assumption 2,*

$$\mathbb{E}[D_M(\pi)] \le \sum_{i:\Delta_i(M)>\lambda} \tilde{S}_i + \lambda T.$$

Proposition 1 shows that, under cumulative control of benchmark jumps, the delay regret is bounded by the same certification-type quantity that appears in the main regret analysis, up to the truncation term λT. Hence the delayed admission of within-round arrivals does not create a new leading-order source of statistical difficulty. More broadly, the result clarifies how the dynamic benchmark enters the regret analysis: the regret of MAB-AA can be viewed as consisting of a principal identification term and an additional delay term induced by benchmark improvement. Under Assumption 2, the latter remains controlled, so benchmark movement acts as a structured perturbation of the elimination problem rather than as a fundamentally separate source of regret.

3.5 General regret bound

We now return to the original dynamic regret $R_{T_0}(\pi) = \sum_{t=1}^{T_0(\pi)} (r^*(t) - r_{I_t})$. By the decomposition in Section 3.2,

$$R_{T_0}(\pi) = R_M(\pi) + D_M(\pi).$$

Theorem 1 controls the principal term $R_M(\pi)$, and Proposition 1 controls the delay term $D_M(\pi)$. Combining the two results yields the following bound for the original regret criterion.

Theorem 2 (General regret bound)**.** *Let $\lambda \ge 2^{-M+1}$. Suppose that Assumptions 1 and 2 hold. Then the dynamic regret of UCB-AA satisfies*

$$\mathbb{E}R_{T_0}(\pi) \le \sum_{i:\Delta_i(M)\ge\lambda} \left(2\tilde{S}_i + \frac{4\big(K(m_i)+5\big)R_i}{N\tilde{\Delta}^2_{\tilde{m}_i}} \right) + 2\lambda T.$$

In particular, under the conservative calibration of N in Corollary 1,

$$\mathbb{E}R_{T_0}(\pi) \lesssim \sqrt{K \log K \cdot T}.$$

3.6 Online UCB-AA

We briefly discuss the case in which the terminal round M is unknown. To remove the fixed-horizon requirement, we use a geometric-restart scheme and run UCB-AA over successive stages with geometrically increasing stage lengths. This yields an anytime version of the procedure, referred to as Online UCB-AA. The idea is standard: each stage is long enough to cover the possibility that the effective horizon is of that order, and restarting across stages incurs only a logarithmic overhead relative to the fixed-M analysis. We retain the algorithm here for completeness and defer the more detailed stage-wise analysis to the Supplementary Material.

Algorithm 2 Online UCB-AA

Input: Base algorithm UCB-AA

Initialize stage index $l \leftarrow 1$
while interaction continues **do**
  $M_l \leftarrow 2^l$, $\tilde{M}_l \leftarrow \sum_{i=1}^{l} M_i$, $\tilde{\Delta}_{M_l} \leftarrow 2^{-M_l}$
  $n_{M_l}(N_l) := \left\lceil \frac{2\log(N_l \tilde{\Delta}_{M_l}^2)}{\tilde{\Delta}_{M_l}^2} \right\rceil$
  Compute N_l from $N_l = 10K(\tilde{M}_l) n_{M_l}(N_l)$ (approx.)
  Run UCB-AA(M_l, N_l) for M_l rounds
  $l \leftarrow l + 1$
end while

At each restart, Online UCB-AA reinitializes the internal state of the base procedure for the new stage, while all pulls across stages are still counted in the cumulative regret. The next corollary summarizes the resulting regret guarantee in the unknown-horizon setting.

Corollary 2. *Suppose that Assumption 1 holds. The expected roundwise regret of Online UCB-AA satisfies*

$$\mathbb{E}R_M(\pi) \lesssim \sqrt{K(M)\, T\, \log T}.$$

If a stagewise version of Assumption 2 holds uniformly across the restart stages, then

$$\mathbb{E}R_{T_0}(\pi) \lesssim \sqrt{K(M)\, T\, \log T}.$$

Corollary 2 shows that the online extension preserves the same square-root-type behavior as the fixed-M procedure, at the cost of only a logarithmic overhead due to geometric restarts. Thus, lack of prior knowledge of the terminal round does not alter the qualitative statistical difficulty of the arriving-arm problem. A more detailed stage-wise bound is given in the Supplementary Material.

4 Lower Bounds in Representative Regimes

As mentioned earlier, without structural restrictions on the arrival process, sublinear regret may be unattainable (Ghalme et al., 2021). A fully general lower bound for the unrestricted arriving-arm model is therefore not our goal here. Instead, we focus on three representative regimes

that isolate distinct hardness mechanisms in MAB-AA: static identification, repeated distractor certification, and near-tied benchmark updates under a drifting comparator.

The statistical hardness of MAB-AA is therefore not monolithic. In the static limit, the problem reduces to the classical identification barrier of stochastic bandits. When a globally optimal incumbent is present from the outset and all late arrivals are inferior, the dominant cost is repeated certification of distractors. When the benchmark improves through a sequence of near-tied arrivals, the difficulty becomes a succession of local tests under a drifting comparator. This section isolates these three mechanisms and relates them to the upper-bound structure in Section 3.

To avoid repeating regime-specific growth conditions, we use the following scale-sensitive extension of the usual notion of uniform goodness.

Definition 1 (Uniform goodness at scale $h(T)$)**.** *Let $\mathcal{V}_T$ be a family of bandit instances indexed by T, and let $h(T) \to \infty$. A sequence of policies $\{\pi^{(T)}\}_{T\geq 1}$ is* uniformly good on $\{\mathcal{V}_T\}$ at scale $h(T)$ *if, for every $a > 0$,*

$$\sup_{\nu\in\mathcal{V}_T} \mathbb{E}_\nu[R_T(\pi^{(T)};\nu)] = o\big(h(T)^a\big) \qquad \text{as } T \to \infty.$$

When $h(T) = T$, it reduces to the classical requirement considered in Lai and Robbins (1985) and Burnetas and Katehakis (1996). For related information-theoretic change-of-measure techniques in best-arm identification, see also Kaufmann, Cappé, and Garivier (2016).

4.1 Static Reduction

We first consider the degenerate no-arrival case. Here $h(T) = T$, so Definition 1 reduces to the usual notion of uniform goodness. The purpose of this subsection is mainly calibrational: when no new arms enter, MAB-AA reduces to the classical stochastic bandit problem, so the arriving-arm framework should recover the familiar logarithmic identification barrier. Accordingly, we restate a standard lower-bound implication from the classical bandit literature (Lai and Robbins, 1985; Burnetas and Katehakis, 1996) in a form adapted to our notation. This result is included for comparison with our arriving-arm analysis, rather than as a new contribution. To avoid unnecessary generality, we use a local formulation under a quadratic KL condition, which is sufficient for our comparison with the upper bounds in Section 3.

Consider a static bandit family with rewards supported on $[0, 1]$. Let $\nu^{(0)}$ be a baseline instance in which arm 0 is the unique optimal arm with mean $r^\star \in (0, 1)$. For $i = 1, \dots, K$, let arm i have reward law P_i and mean $r_i = r^\star - \Delta_i$ where $\Delta_i > 0$.

Theorem 3 (Static lower bound)**.** *Assume that for each $i = 1, \dots, K$ there exists a distribution $Q_i \in \mathcal{P}([0, 1])$ such that $\eta_i := \mathbb{E}_{Q_i}[X] - r^\star > 0$ and $D(P_i\|Q_i) \leq C_i\Delta_i^2$ for some finite constant $C_i > 0$. Let $\nu^{(i)}$ denote the instance obtained from $\nu^{(0)}$ by replacing only the reward law of arm i from P_i to Q_i. If a policy π is uniformly good on the local family $\mathcal{V} := \{\nu^{(0)}, \nu^{(1)}, \dots, \nu^{(K)}\}$, then*

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi;\nu^{(0)})] \geq c \sum_{i=1}^{K} \frac{\log T}{C_i\Delta_i}$$

for all sufficiently large T, where $c > 0$ is an absolute constant. In particular, if $\sup_{1\leq i\leq K} C_i \leq$

$C_\star < \infty$, *then*

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi;\nu^{(0)})] \geq c\, C_\star^{-1} \sum_{i=1}^{K} \frac{\log T}{\Delta_i}$$

for all sufficiently large T.

Example 1 (Static specialization)**.** *In the absence of arrivals, UCB-AA specializes to an elimination-style stochastic bandit procedure and achieves* $\tilde{O}\left(\sum_i \frac{\log(T\Delta_i^2)}{\Delta_i}\right)$ *instance-dependent regret. Together with Theorem 3, this shows that the arriving-arm reduction does not sacrifice statistical efficiency in the static limit, up to logarithmic refinements.*

4.2 Sequential Distractors

We next consider a regime in which a globally optimal incumbent is present from the outset, while every newly arriving arm is suboptimal under the baseline instance. The difficulty is therefore repeated certification: each arrival creates a local alternative under which that arm would become optimal, so the learner must accumulate logarithmic evidence before discarding it.

For each horizon T, let $\nu^{(0)}$ be a baseline instance in which arm 0 is the unique optimal arm, available from time 1, with mean $r^\star \in (0,1)$. Let $\mathcal{I}_T$ be a collection of arriving distractor arms. For each $i \in \mathcal{I}_T$, suppose arm i arrives at time $\tau_i \leq T$, has reward law P_i, and mean $r_i = r^\star - \Delta_i$ where $\Delta_i > 0$. Write

$$H_i(T) := T - \tau_i + 1, \qquad H_{\min}(T) := \min_{i \in \mathcal{I}_T} H_i(T),$$

and assume $H_{\min}(T) \to \infty$. In this regime we evaluate uniform goodness at the effective scale $h(T) = H_{\min}(T)$.

Theorem 4 (Distractor lower bound under local alternatives)**.** *Suppose that for each* $i \in \mathcal{I}_T$ *there exists a law* $Q_i \in \mathcal{P}([0,1])$ *such that* $\eta_i := \mathbb{E}_{Q_i}[X] - r^\star > 0$ *and* $d_i := D(P_i \| Q_i) < \infty$. *Let* $\nu^{(i)}$ *denote the instance obtained from* $\nu^{(0)}$ *by replacing the reward law of arm* i *from* P_i *to* Q_i, *and set* $\mathcal{V}_T := \{\nu^{(0)}\} \cup \{\nu^{(i)} : i \in \mathcal{I}_T\}$. *Assume that, for all sufficiently large* T,

$$\inf_{i \in \mathcal{I}_T} \Delta_i \geq \underline{\Delta} > 0, \qquad \inf_{i \in \mathcal{I}_T} \eta_i \geq \underline{\eta} > 0, \qquad \sup_{i \in \mathcal{I}_T} d_i \leq \overline{d} < \infty.$$

Then any sequence $\{\pi^{(T)}\}_{T \geq 1}$ *uniformly good on* $\mathcal{V}_T$ *satisfies*

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)};\nu^{(0)})] \geq c \sum_{i \in \mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_i(T) \geq c\, \underline{\Delta}\, \overline{d}^{-1} \, |\mathcal{I}_T| \, \log H_{\min}(T)$$

for all sufficiently large T, *where* $c > 0$ *is an absolute constant.*

Corollary 3 (Two canonical distractor regimes)**.** *Under the assumptions of Theorem 4, suppose in addition that one of the following holds:*

(i) Equal-distance arrival: *each arm in* $\mathcal{I}_T = \{1, \ldots, K_T\}$ *has the same arrival interval* $H := \lfloor T/K_T \rfloor$ *which means* $\tau_i = (i-1)H + 1$,*and* H *tends to infinity.*

(ii) Early arrival: *there exists a fixed* $\alpha \in (0,1)$ *such that* $\tau_i \le \alpha T$ *for every* $i \in \mathcal{I}_T$.

Then

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)};\nu^{(0)})] \ge c \sum_{i\in\mathcal{I}_T} \frac{\Delta_i}{d_i} \log H_{\min}(T)$$

for all sufficiently large T. *In particular, if*

$$\sup_{i\in\mathcal{I}_T} d_i \le \overline{d} < \infty, \qquad \inf_{i\in\mathcal{I}_T} \Delta_i \ge \underline{\Delta} > 0, \qquad |\mathcal{I}_T| \asymp K_T,$$

then

$$\mathbb{E}_{\nu^{(0)}}[R_T(\pi^{(T)};\nu^{(0)})] = \Omega\big(K_T \log H_{\min}(T)\big).$$

The lower bound reflects a repeated certification cost: although the arriving arms are inferior in the baseline instance, a uniformly good policy cannot ignore them, because each arrival induces a local alternative under which that arm is optimal.

Example 2 (Fixed-gap distractors)**.** *Consider an initial set of arms* $A_0 = \{*, 0\}$, *where arm* $*$ *is globally optimal with mean* $\mu_* = \frac{3}{4}$ *and arm* 0 *has mean* $\mu_0 = \frac{1}{4}$. *For each round* $m = 1, \dots, M$, *a single new arm* $i = m$ *arrives with* $\mu_i = \frac{1}{2} - 2^{-i-2}$. *Let* $K(m) = m+2$ *denote an upper bound on the number of arms available up to round* m. *All new arrivals are distractors: their means converge to* $1/2$, *while the optimal arm remains fixed at* $3/4$, *so the gaps stay bounded away from zero. Applying Theorem 1 with* $N = 10K(M)n_M$ *yields* $\mathbb{E}R_T(\pi_{\text{UCB-AA}}) = \mathbb{E}R_M(\pi_{\text{UCB-AA}}) = \widetilde{O}(K(M)\log T)$.

Combined with Theorem 4, this gives the comparison

$$\widetilde{O}(K_T \log T) \qquad \text{versus} \qquad \Omega(K_T \log(T/K_T)),$$

so in the canonical sparse-arrival regime $K_T = o(T)$ the rates agree up to logarithmic factors. A complementary failure-mode construction showing that the absence of pre-elimination can lead to linear regret is deferred to the Supplementary Material.

4.3 Progressive Near-tied Arrivals

We finally consider a regime in which the benchmark improves through a sequence of near-tied arrivals. Unlike the distractor setting, the main difficulty is no longer certifying clearly inferior arms, but repeatedly deciding whether a new challenger lies slightly below or slightly above the current reference level. Because this mechanism is intrinsically local and horizon dependent, the natural lower bound here is a finite-horizon minimax statement.

Fix a horizon T and integers $K_T \ge 1$, $H_1, \dots, H_{K_T} \in \mathbb{N}$ such that $\sum_{m=1}^{K_T} H_m = T$. Define checkpoints and blocks by

$$s_0 := 0, \qquad s_m := \sum_{\ell=1}^{m} H_\ell, \qquad \mathcal{B}_m := \{s_{m-1}+1, \dots, s_m\}, \qquad m = 1, \dots, K_T.$$

Let $\gamma \in (0, 1/2]$, and choose $\theta_1, \dots, \theta_{K_T} \in (0,1)$, $\varepsilon_1, \dots, \varepsilon_{K_T} > 0$ such that

$$\theta_m - \varepsilon_m,\ \theta_m,\ \theta_m + \varepsilon_m \in [\gamma, 1-\gamma], \qquad m = 1, \dots, K_T, \tag{3}$$

$$\theta_{m+1} \geq \theta_m + \varepsilon_m, \qquad m = 1, \ldots, K_T - 1. \tag{4}$$

For each sign vector $\sigma = (\sigma_1, \ldots, \sigma_{K_T}) \in \{-1, +1\}^{K_T}$, define an arriving-arm instance ν_σ in which, at time $s_{m-1} + 1$, two Bernoulli arms c_m, d_m arrive and remain available thereafter, with means $\mu(c_m) = \theta_m$ and $\mu(d_m) = \theta_m + \sigma_m \varepsilon_m$. Let $\mathcal{V}_T^{\mathrm{prog}} := \{\nu_\sigma : \sigma \in \{-1, +1\}^{K_T}\}$.

Theorem 5 (Deterministic-checkpoint minimax lower bound)**.** *Under (3)–(4), every non-anticipating policy π satisfies*

$$\sup_{\nu \in \mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu[R_T(\pi; \nu)] \geq \frac{1}{2} \sum_{m=1}^{K_T} H_m \varepsilon_m \left(1 - \sqrt{\frac{2 H_m \varepsilon_m^2}{\gamma(1-\gamma)}}\right)_+. \tag{5}$$

In particular, if there exists a constant $\kappa \in (0, 1/2)$ such that

$$H_m \varepsilon_m^2 \leq \kappa\, \gamma(1-\gamma), \qquad m = 1, \ldots, K_T,$$

$$\textit{then} \quad \sup_{\nu \in \mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu[R_T(\pi; \nu)] \geq c_\kappa \sum_{m=1}^{K_T} H_m \varepsilon_m, \qquad c_\kappa := \frac{1}{2}(1 - \sqrt{2\kappa}) > 0. \tag{6}$$

The same lower bounds remain valid after taking $\inf_\pi$ on the left-hand side.

Each block creates a fresh one-sided local test: the learner must determine whether the new challenger lies slightly below or slightly above the current benchmark. When $H_m \varepsilon_m^2$ is of constant order, the resulting regret contribution is of order $H_m \varepsilon_m$, and these local-testing costs add across blocks.

Corollary 4 ($\sqrt{K_T T}$ lower bound under near-homogeneous local tests)**.** *Under the assumptions of Theorem 5, assume in addition that there exist constants $c_H, C_H, c_\varepsilon, C_\varepsilon > 0$ and $\kappa \in (0, 1/2)$ such that, for every $m = 1, \ldots, K_T$,*

$$c_H \frac{T}{K_T} \leq H_m \leq C_H \frac{T}{K_T}, \qquad c_\varepsilon \sqrt{\frac{K_T}{T}} \leq \varepsilon_m \leq C_\varepsilon \sqrt{\frac{K_T}{T}},$$

and $C_H C_\varepsilon^2 \leq \kappa\, \gamma(1-\gamma)$. Then

$$\inf_\pi \sup_{\nu \in \mathcal{V}_T^{\mathrm{prog}}} \mathbb{E}_\nu[R_T(\pi; \nu)] \geq c \sqrt{K_T T}$$

for all sufficiently large T, where $c > 0$ depends only on c_H, c_ε, and κ.

Corollary 4 identifies the canonical square-root minimax scale in a near-homogeneous local-testing regime. Our general upper bound in Section 3 matches this dependence up to logarithmic factors. The example below is not a literal specialization of the deterministic-checkpoint lower bound, because UCB-AA operates with endogenous random checkpoints, but it exhibits the same qualitative mechanism.

Example 3 (Monotone near-tied ladder)**.** *Fix $M \geq 5$ and define $\xi_m := \frac{1}{2} - 2^{-m-2}$ for $m = 0, 1, \ldots, M$. At round 0, there is a single arm a_0, so that $A_0 = \{a_0\}$ and $\mu(a_0) = \xi_0$. For each round $m = 1, \ldots, M$, two new Bernoulli arms arrive, namely $A_m = \{c_m, d_m\}$, with means $\mu(c_m) = \xi_{m-1}$ and $\mu(d_m) = \xi_m$. Hence the unique optimal arm in round m is d_m, with $r^*(m) = \xi_m$, and the cumulative number of available arms is $K(m) = |A(m)| = 2m + 1$. Choosing $N = 10K(M)n_M$ yields $\mathbb{E}R_T(\pi_{\text{UCB-AA}}) = \mathbb{E}R_M(\pi_{\text{UCB-AA}}) = O(\sqrt{T \log K(M)})$.*

Example 3 should be interpreted as an endogenous-checkpoint illustration of the same local-testing mechanism. The deterministic-checkpoint lower bound gives the canonical minimax benchmark, while the UCB-AA analysis yields the same square-root-type behavior up to an additional logarithmic factor along its own random checkpoint schedule.

Taken together, these lower-bound examples show that MAB-AA contains several distinct sources of statistical difficulty. They are not intended as term-by-term converses to the upper bounds in Section 3; rather, they clarify why the analysis must separately control the costs of identifying good incumbent arms, certifying arriving competitors, and tracking benchmark changes.

5 Simulation

We evaluate UCB-AA on two synthetic arriving-arm environments. The first contains abrupt improvements in the best available arm, while the second contains frequent near-optimal arrivals. Both settings use horizon $T = 60000$. We report three metrics:

- Cumulative dynamic regret R_t: measures loss relative to the best currently available arm, $R_t = \sum_{s=1}^{t}(r^*(s) - r_{I_s})$, where $r^*(s) = \max_{i \in \mathcal{A}_s} r_i$ and I_s is the arm pulled at time s.
- Wasted pulls W_T: measures exploration spent on clearly suboptimal choices, $W_T = \sum_{s=1}^{T} \mathbb{I}\{\Delta_{I_s}(s) \geq 0.1\}$.
- Active-set size: measures the pruning effect of elimination, defined as the number of arms retained after elimination steps.

We consider two distinct arrival scenarios designed to stress-test different challenges inherent to the MAB-AA setting; for specific distribution parameter settings, refer to the supplementary materials:

Scenario 1 (Progressive Step-up Arrivals)

The environment starts with $K_0 = 10$ arms and background arrivals following a Poisson process with rate $\lambda = 1/200$, with means drawn from a bi-modal mixture. To create benchmark shifts, high-performing step arms are introduced at $t/T \in \{0.3, 0.6, 0.9\}$, with $\mu_k = \min(1, r_0 + k\Delta_{\text{step}})$, $r_0 = 0.6$, and $\Delta_{\text{step}} = 0.05$. This setting tests whether an algorithm can quickly detect and switch to newly arrived superior arms.

Scenario 2 (Near-tied Top Set)

The environment starts with $K_0 = 20$ arms, including $m_0 = 5$ near-optimal arms with means from Uniform$(0.7, 0.8)$. New arms arrive according to a Bernoulli process with probability $\lambda = 1/200$, and 30% of them are near-optimal. This setting represents mature decision problems with many near-tied top arms, where small performance differences must be detected without excessive exploration.

To ensure a fair comparison, we adapt classical algorithms to handle the continuous arrival of new arms. Their implementations are described below; additional details are provided in the supplementary materials.

1. UCB-AA (Proposed): the proposed method, which buffers newly arrived arms and screens them through pre-elimination before full competition. We set $N = e/\tilde{\Delta}_M^2$ in the simulations.

2. Growing UCB1 (G-UCB1): UCB1 (Auer, Cesa-Bianchi, and Fischer, 2002) with immediate admission of new arms, using index $\hat{\mu}_{i,n} + \sqrt{2 \ln t / n}$.

3. Growing MOSS (G-MOSS): MOSS (Audibert and Bubeck, 2009) with immediate admission of new arms, using index $\hat{\mu}_i + \sqrt{\max\{0, \log(T/(K_t n_i))\}/n_i}$, where $K_t = |\mathcal{A}_t|$.

4. Dynamic Successive Elimination (D-SE): an SE-style baseline (Perchet and Rigollet, 2013) that admits new arms at each round, samples them to match incumbent pull counts, and then applies the SE elimination rule.

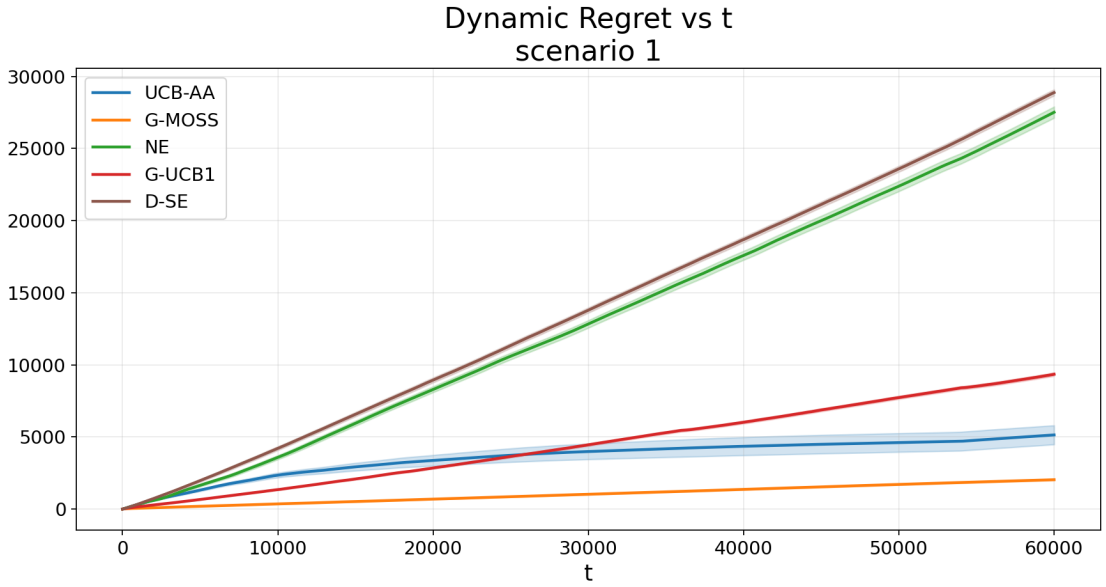
5. Naive Elimination (NE): UCB-AA with the same formal elimination rule but without the pre-elimination stage.

For each scenario, we ran 100 independent simulations. The figures below report the average cumulative dynamic regret and wasted pulls, with 95% confidence intervals.

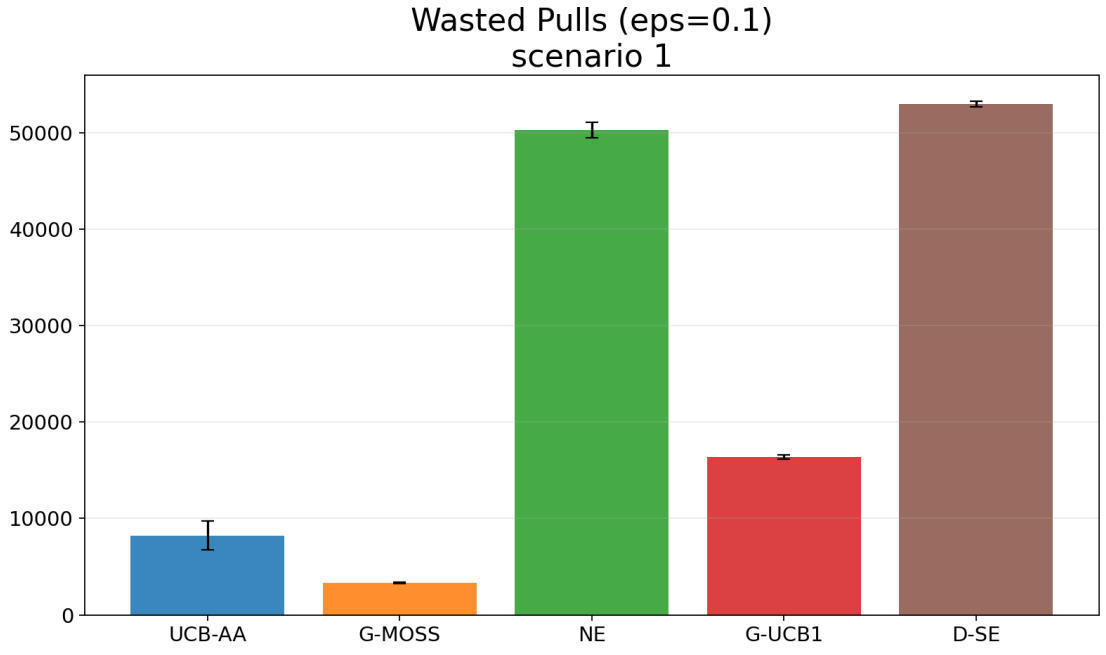
The simulation results show that UCB-AA substantially reduces the number of wasted pulls (W_T) relative to Growing UCB1 and the elimination-based baselines. The main reason is that newly arriving arms are not allowed to compete immediately on equal footing with well-sampled incumbents. Instead, UCB-AA uses a buffered pre-elimination step to screen out clearly inferior late arrivals before full competition, thereby protecting the sampling budget of the currently most competitive arms. By contrast, Growing UCB1 continues to allocate exploration to each new arrival through its uncertainty bonus, while Naive Elimination and Dynamic SE treat new and old arms too symmetrically at entry, leading to larger regret.

A second advantage of UCB-AA is that it maintains a compact active set whose members have survived repeated statistical comparisons over time. Methods such as Growing UCB1 and Growing MOSS do not explicitly prune the arm set once arms arrive, whereas UCB-AA actively shrinks its working pool $|\mathcal{C}_t|$ through pre-elimination and formal elimination. As a result, UCB-AA not only controls regret well, but also returns a smaller set of arms that have survived repeated statistical comparisons and are therefore more likely to be genuinely competitive.

At the same time, UCB-AA does not uniformly dominate Growing MOSS in cumulative dynamic regret (R_T); instead, their regret levels are often comparable, and Growing MOSS can be

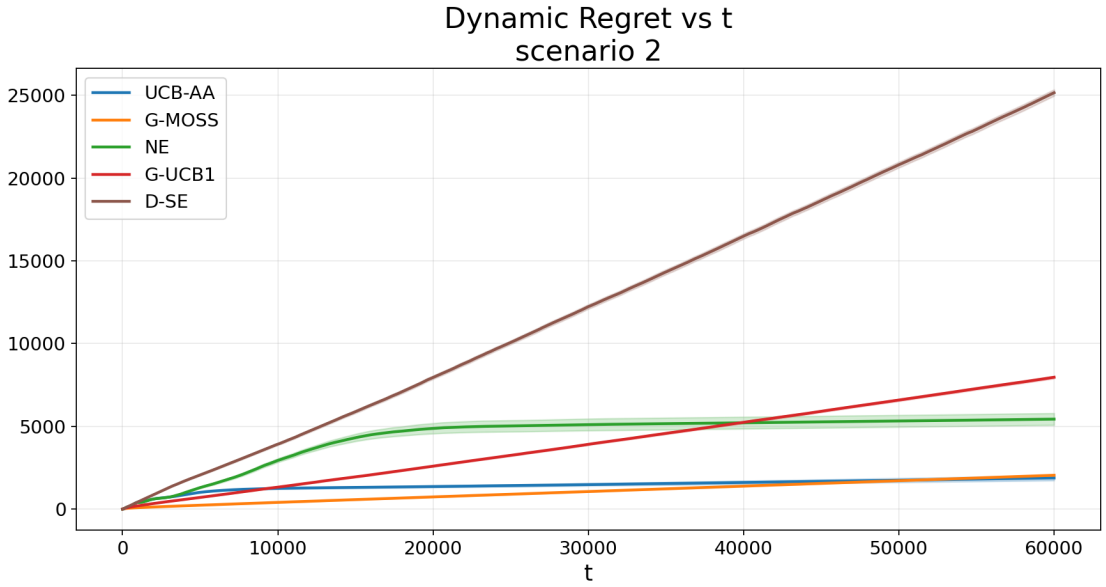


(a) Cumulative dynamic regret (Scenario 1)

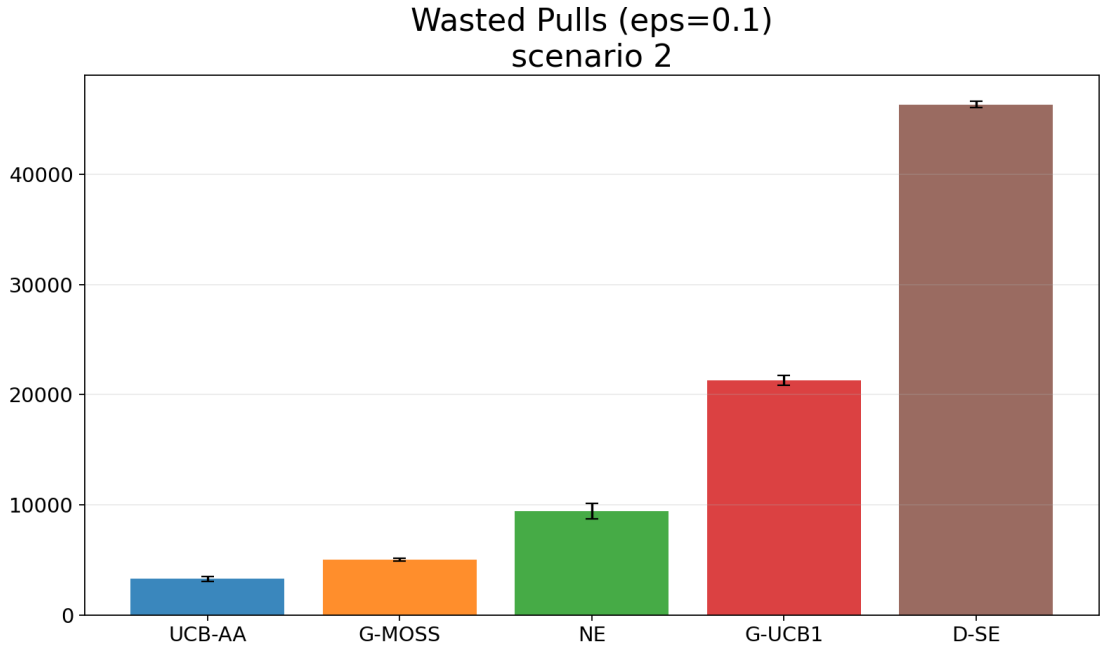


(b) Wasted pulls (Scenario 1)

Figure 2: Experimental results for Scenario 1 (Progressive step-up arrivals). Panel (a) shows cumulative dynamic regret, and panel (b) compares wasted pulls (W_T) and the final number of active arms. Results are averaged over 100 independent runs.

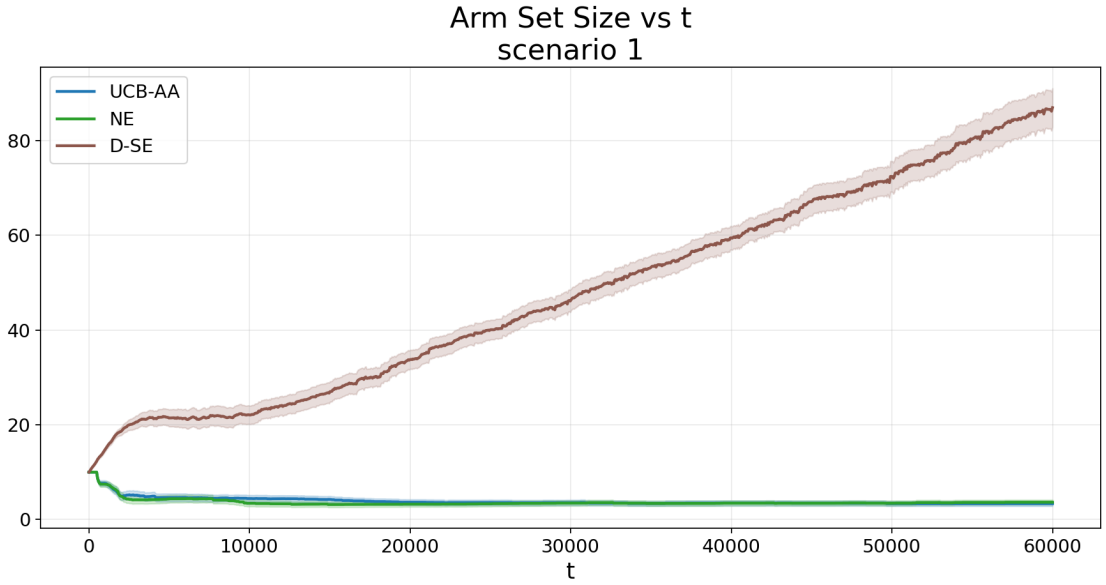


(a) Cumulative dynamic regret (Scenario 2)

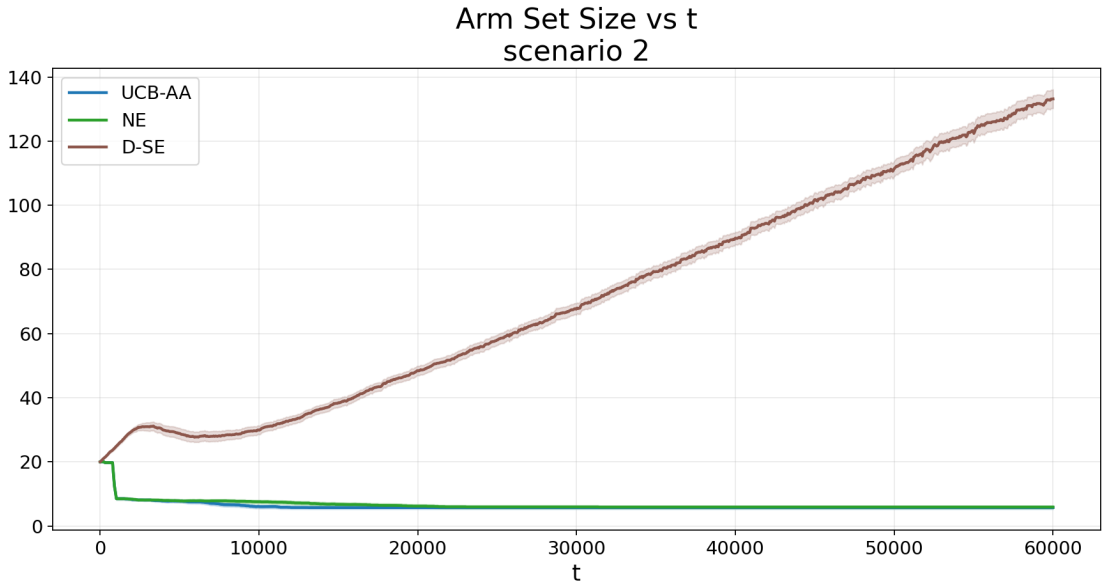


(b) Wasted pulls (Scenario 2)

Figure 3: Experimental results for Scenario 2 (Near-tied top set). UCB-AA's explicit elimination mechanism maintains a compact decision space while achieving regret performance comparable to Growing MOSS.



(a) Active arm size (Scenario 1)



(b) Active arm size (Scenario 2)

Figure 4: Active arm size for two scenarios. UCB-AA's explicit elimination mechanism maintains a compact decision space while achieving regret performance comparable to Growing MOSS.

slightly better in settings where many arms are near-tied. This reflects the cost of explicit hard elimination. In UCB-AA, an arm is removed only when its upper confidence bound falls below the largest lower confidence bound among the surviving arms, $UCB_j < \max_i LCB_i$. Because this rule requires strong evidence, UCB-AA may spend additional samples to certify that a borderline arm is truly inferior before discarding it. By contrast, Growing MOSS performs only a soft form of elimination: weak arms receive fewer pulls when their indices fall behind, but they remain in the available set and can be sampled again if their uncertainty remains large.

This trade-off also indicates when hard elimination is useful. If the only objective is cumulative regret over a moderate horizon and keeping all arrived arms active is essentially free, then soft index-based methods such as Growing MOSS can be highly competitive. However, in many arriving-arm applications, the active set itself is costly to maintain. Examples include recommendation systems with continuously added items, adaptive A/B testing with many newly launched designs, biomedical screening or dose-finding studies with costly or risky candidates, and sequential model or policy selection where each surviving candidate requires repeated evaluation. In such settings, reducing the active set is not merely a computational convenience; it is part of the statistical objective. UCB-AA therefore trades a modest amount of additional exploration for certified pruning of clearly inferior late arrivals, thereby controlling both wasted pulls and the operational burden created by an expanding action space.

6 Discussion

We studied stochastic multi-armed bandits with arriving arms, where the available action set expands over time and the performance is measured against the best currently available arm. The distinct features of arrival information discrepancy (AID) and drifting benchmark (DB) depart from the classical fixed-oracle benchmark and creates two coupled statistical challenges: repeated certification of newly arriving competitors and a moving target to catch up.

Our main contribution shows that these challenges can be handled in a structured way. The round-based reduction separates the principal certification burden from the additional delay induced by benchmark movement, while UCB-AA uses a pre-elimination stage to screen late arrivals before they enter full competition. Under suitable control of gap evolution, this yields sublinear regret together with a compact active set. The lower-bound regimes further clarify that the hardness of MAB-AA has multiple sources: the static regime recovers the classical logarithmic identification barrier, distractor arrivals induce cumulative certification costs, and near-tied improvements create a local-testing difficulty under a drifting benchmark.

Several extensions are natural. One direction is to incorporate contextual information or arm-level covariates when available, which may improve reward via individualization as well as the screening efficiency for newly arriving arms. Another direction is to study settings with post-arrival reward drift, while retaining the central focus on how an expanding action set affects exploration and regret. Finally, the arriving-arm framework can be extended to settings involving delayed feedback, operational resource constraints, and arrival processes calibrated from real application data.

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